

Security Risk Assessment and Network Hardening Recommendations

1. Scenario

Conducted a security risk assessment for a social media organization following a data breach caused by unmanaged vulnerabilities.

2. Responsibilities

- identified security weakness contributing to the breach
- Evaluated network hardening methods
- Assessed effectiveness of preventive security controls
- Recommended security measures to reduce future risk exposure
- Documented mitigation strategy and security improvements

3. Recommended controls

A) MFA

- reduced risk of credential compromise
- Added an additional authentication layer

B) Strong password policy

- password complexity requirements
- Password history enforcement
- Account lockout thresholds
- Protection against brute-force attacks

C) Firewall maintenance

- regular firewall rule reviews
- Removal of outdated configurations
- Blocking suspicious traffic patterns
- Improved protection against DoS and DDoS attacks.

4. Skills demonstrated

- security risk assessment
- Security control selection
- Risk mitigation planning
- Network hardening
- Security governance
- Cybersecurity documentation
- Security recommendations